

Name: SOLUTION MANUAL

MATH 241

QUIZ #6

SUMMER 2014 8 minutes x 2.5 = 20 minutes!

Problem 1. (5 Points) 7:51-7:54 3 minutes.

Given the difference equation below

$x[k] = \frac{5}{6}x[k-1] - \frac{1}{6}x[k-2]$ with the initial conditions $x[0] = x[-1] = -1$. Find the general expression for $x[k]$ for $k \geq 1$, using any method you prefer.

Characteristic equation:

$$\left(\lambda^2 - \frac{5}{6}\lambda + \frac{1}{6}\right) = \left(\lambda - \frac{1}{2}\right)\left(\lambda - \frac{1}{3}\right) \Rightarrow x[n] = \alpha_1 \lambda_1^n + \alpha_2 \lambda_2^n = \alpha_1 \left(\frac{1}{2}\right)^n + \alpha_2 \left(\frac{1}{3}\right)^n$$

$\lambda_1 = \frac{1}{2} \quad \lambda_2 = \frac{1}{3}$

$$\begin{cases} x[0] = -1 = \alpha_1 + \alpha_2 \\ x[-1] = -1 = 2\alpha_1 + 3\alpha_2 \end{cases} \quad \begin{aligned} \alpha_2 &= 2 = -1 & \alpha_2 &= 1 \\ \alpha_1 &= -1 - \alpha_2 = -2 \end{aligned}$$

$$x[n] = \left(\frac{1}{3}\right)^n - 2\left(\frac{1}{2}\right)^n$$

$$x[1] = \frac{1}{3} = 1 = -\frac{2}{3} = \frac{5}{6}(-1) - \frac{1}{6}(-1) = -\frac{5}{6} + \frac{1}{6} = -\frac{4}{6} \quad \checkmark$$

Problem 2. (5 Points) 7:54 - 7:59 → 5 minutes

Given the difference equation below,

$$x[k] = x[k-1] - \frac{1}{4}x[k-2] \text{ with the initial conditions } x[-1]=1, x[-2]=3.$$

Find the general expression of $x[k]$ for $k \geq 0$, using the Z-transform.

$$X(z) = z^{-1}X(z) + x[-1] - \frac{1}{4} \left(z^{-2}X(z) + z^{-1}x[-1] + x[-2] \right)$$

$$X(z) \left(1 - z^{-1} + \frac{1}{4}z^{-2} \right) = x[-1] - \frac{1}{4}x[-2] - z^{-1} \frac{x[-1]}{4}$$

$$X(z) = \frac{\frac{1}{4}(1 - z^{-1})}{\left(1 - \frac{1}{2}z^{-1}\right)^2} = \frac{1}{4} \frac{\left(1 - \frac{1}{2}z^{-1}\right) - \frac{1}{2}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)^2} = \frac{1}{4} \left(\frac{1}{1 - \frac{1}{2}z^{-1}} - \frac{\frac{1}{2}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)^2} \right)$$

$$\xrightarrow{Z^{-1}} \frac{1}{4} \left(\left(\frac{1}{2}\right)^n - n \left(\frac{1}{2}\right)^n \right)$$

$$x[n] = \frac{1}{4} \left(1 - n \right) \left(\frac{1}{2} \right)^n = (1-n) \left(\frac{1}{2} \right)^{n+2}$$

$$x[-1] = 2 \left(\frac{1}{2} \right)^{2-1} = 1 \quad x[-2] = (1+2) \left(\frac{1}{2} \right)^{2-2} = 3 \quad \checkmark$$

$$x[0] = \frac{1}{4} = 1 - \frac{3}{4} \quad \checkmark$$